

DEPI React Project

Team Members:

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Group:

GIZ4_SWD2_S6_DEPI4

Project Name:

AI-Powered Developer Portfolio Builder

GitHub Repo:

<https://github.com/Basel-Ahmed-TECH/DEPI-React-GP>

1. Software Requirements Specification (SRS)

1.1 Introduction

PortfolioGenie is an AI-powered web application designed to assist junior developers in creating professional portfolio websites efficiently. The system integrates with the GitHub API to retrieve user repository data and utilizes AI techniques (prompt engineering and LLMs) to generate optimized content such as project descriptions, skills summaries, and an “About Me” section.

1.2 Objectives

- Enable developers to build professional portfolios
- Generate AI-enhanced and SEO-friendly content
- Provide a real-time, interactive portfolio preview
- Export clean, maintainable, and deployment-ready code

1.3 Scope

The system includes the following functionalities:

- GitHub repository analysis and data extraction
- Multi-step portfolio builder interface
- AI-powered content generation and optimization
- Real-time portfolio preview with responsive design
- Export of ready-to-deploy portfolio code
- Support for multiple portfolio templates

1.4 Functional Requirements

◆ GitHub Integration

- Retrieve user repositories using GitHub API
- Extract repository metadata (languages, stars, descriptions)

◆ Portfolio Builder

- Multi-step form for user input:
 - Personal information
 - Project selection
 - Skills and technologies
- Allow user customization

◆ AI Content Generation

- Generate:
 - Project descriptions
 - Skills summary
 - Professional “About Me” section
- Improve readability, tone, and SEO

◆ Live Preview

- Display portfolio in real time
- Support responsive layouts (mobile, tablet, desktop)

◆ Template System

- Provide multiple portfolio templates (3–5 designs)
- Allow users to switch between templates dynamically

◆ Export System

- Export portfolio as ZIP file
- Generate clean HTML/CSS/JS or React code

1.5 Non-Functional Requirements

- **Performance:** Fast loading and response time
- **Usability:** Simple and intuitive user interface
- **Responsiveness:** Fully responsive design across devices
- **Scalability:** Ability to handle multiple users
- **Security:** Secure API communication and data handling

2. Software Design Document (SDD)

2.1 Architecture

- **Frontend:** React.js
- **API Integration:** GitHub REST API
- **AI Layer:** Prompt Engineering + LLM-based content generation

2.2 System Modules

- ◆ GitHub Integration Module
 - Fetch repositories from GitHub
 - Extract and process repository data
- ◆ Portfolio Builder Module
 - Multi-step user interface
 - State management using React Hooks / Context API
- ◆ AI Content Optimization Module
 - Prompt templates for content generation
 - Generate and refine:
 - Project descriptions
 - About section
 - Skills summary

◆ Live Preview Module

- Real-time portfolio rendering
- Responsive preview across devices

◆ Template & Theme Module

- Multiple portfolio templates (3–5)
- Theme customization (colors, layout, typography)

◆ Export Module

- Generate portfolio files
- Download as ZIP
- Provide deployment-ready structure

2.3 Data Flow

1. User enters GitHub username
2. System retrieves repositories via GitHub API
3. Data is processed and structured
4. AI generates optimized content
5. Portfolio is rendered in preview
6. User exports final portfolio